MILITARY SPECIFICATION

PROJECTILE, 152MM: HEAT, M409 METAL PARTS, LESS FLUTED LINER

1. SCOPE

1.1 This specification covers Projectile, 152MM: HEAT, M409 Metal Parts, Less Fluted Liner.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for
MIL-I-6868	- Inspection Process, Magnetic Particle
MIL-R-11073	- Rotating Band Blank, Sintered Iron
MIL-C-60539	- Coating, Anodic, Conventional, for Aluminum and Aluminum Alloys

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-410	- Certification of Magnetic Particle Inspection Personnel

FSC 1320

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2.1 (Cont'd)

MIL-STD-171 - Finishing of Metal and Wood Surfaces
MIL-STD-1167 - Ammunition Data Card
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

Munitions Command
F10534407 - Projectile, 152MM: HEAT, M409 Metal Parts
D10534415 - Fluted Liner
F8889328 - Pallet for Fixed and Semifixed Projectiles
LI-10534407 - Index of Inspection Equipment List for Projectile, 152MM: HEAT, M409 Metal Parts

PUBLICATIONS

Picatinny Arsenal
PA-PD-792 - Equipment, Testing or Measuring for Ammunition Items

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ASA-B46.1 - Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, N. Y.)

ASTM-E-8
ASTM-E-18
ASTM-A-317

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications and the following:

3.2 Manufacture of metal parts.

3.2.1 Windshield.-The windshield may be manufactured by any one of the following methods to attain required mechanical properties:

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3.2.1.1 Hot forge - heat treatment.-Windshields manufactured by this method shall be subjected to a heat-treatment to consist of heating above the critical temperature, not to exceed 1600 degrees F., followed by an oil quench. After quenching, the component shall be tempered at a temperature above 700 degrees F. for a time suitable to produce the required mechanical properties. After removal from the tempering furnace, the component shall be cooled in air (forced air permitted) to a temperature of not more than 700 degrees F., after which cooling may be accelerated by a water spray or similar means.

3.2.1.2 Hot forge - normalize.-Windshields manufactured by this method must be re-heated after forging to above the critical temperature, not to exceed 1650 degrees F. in a continuous conveyorized or rotary-hearth type furnace only. Batch-type operations or stacking of components are not permissable. After heating at a suitable temperature and time to meet mechanical properties on cooling, the component shall be cooled in air (forced air permitted) to a temperature of not more than 700 degrees F., after which cooling may be accelerated by a water spray or similar means.

3.2.1.3 Cold shaping.-Cold shaping, followed by stress relief, shall be so accomplished as to produce the required mechanical properties throughout the component part. The component shall be stress relieved for a minimum of one (1) hour at a minimum temperature of 700 degrees F. following the final forming operation during which plastic flow takes place. The stress relief time and temperature at or above the specified minimum shall be such as to produce the required mechanical properties.

3.2.2 Body.-The body shall be manufactured by the hot forge-heat treatment method of manufacture only. The heat treatment process shall be such as to produce the required mechanical properties.

3.2.2.1 Decarburization.-The body after final machining shall have no evidence of decarburization on the outside diameter.

3.2.3 Base plug and fuze lock cup.

3.2.3.1 Heat treatment.-These components shall be subjected to a heat treatment to consist of heating above the critical temperature, not to exceed 1600 degrees F., followed by an oil quench. After quenching, the components shall be tempered at a temperature above 700 degrees F. for a time suitable to produce the required mechanical properties. After removal from the tempering furnace, the components shall be cooled in air (forced air permitted) to a temperature of not more than 700 degrees F., after which cooling may be accelerated by a water spray or similar means. Regardless of the method of forming or the material selected, heat treatment shall be such as to produce the required mechanical properties in the component parts.

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3.2.4 Locking ring.-The locking ring shall be machined from mechanical (seamless) tubing, the mechanical properties of which comply with the requirements of the applicable drawing.

3.2.5 Cleanliness of band seat.-Prior to application of the rotating band blank, the body band seat shall be clean and free of rust, dirt, oil, grease, or corrosive materials.

3.2.6 Protective finish.-Protective finish shall be in accordance with the applicable drawings and specifications.

3.2.7 Painting.-Painting shall comply with the requirements of the applicable drawings. All paint shall be dry to the touch before testing and packing for shipment.

3.3 Proving ground functional tests.-The assembled projectile metal parts shall reveal no evidence of failure when tested at excess and service pressures. Metal parts failure shall be:

a. Deformation of any component attributable to noncompliance to an extent greater than .020 of an inch.

b. Loss or breakup of metal part components (except engraving of rotating band).

3.3.1 Fragmentation.-Initial samples of projectile assemblies shall be tested for fragmentation to evaluate quality of the end item (for information only).

3.4 Workmanship.-The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.4.1 Material defects.-Molded components shall be free of porosity and excessive flash.

3.4.2 Metal defects.-All components shall be free from cracks, splits, porosity, bursts, cold shuts, pipe, folds, seams, or evidence of unsound material other than indications of solid nonmetallic inclusons in the rolling direction of the material to a maximum length of 1/8 inch. (See 6.6.)

3.4.2.1 Body cavity.-In addition to the requirements of 3.4.2 the body cavity shall be free of scale, fins, burrs, draw marks, laminations, imbedded foreign matter, pits and sharp edges. (See 6.6.)

3.4.3 Burr.-No components shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.4.4 Foreign matter.-All parts and assemblies shall be free of chips, dirt, grease, oil, rust or other foreign matter.

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3.4.5 Threads.-Threads shall be full and undamaged for the entire minimum length or depth as specified on the applicable drawings, and shall be in accordance with the provisions of MIL-A-2550.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.-Inspection shall be in accordance with MIL-A-2550, except as specified herein.

4.1.3 Inspection criteria.

a. Successful passage of all inspections required by the drawings and specifications shall be considered as a condition precedent to performing the proving ground functional tests (see 3.3).

b. Acceptance or rejection of a lot by the Government shall be contingent on successful passage of the proving ground functional tests.

4.2 First article sample.

4.2.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of fifty-two projectile metal parts as follows:

37 metal parts assemblies complete, including paint
5 metal parts assemblies complete, except paint
5 sets of unassembled component parts complete, including protective coating except those components requiring protective finish in accordance with MIL-STD-171, finish number 5.1.1 shall be furnished without protective coating
5 sets of subassemblies complete, except the body and band assembly shall be submitted without protective coating

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4.2.1 (Cont'd)

Thirty-two each of the above metal parts assemblies complete shall be forwarded to a designated proving ground for functional tests. The sample shall be manufactured in the same manner using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.-After the inspection and provisional acceptance at source, the sample shall be inspected for all the requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.-Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-1235, as applicable.

4.3.1.2 Heat treatment lots.

4.3.1.2.1 Bodies.-At the start of production, for the first 1000 bodies, a heat treatment lot shall consist of not more than one heat of steel nor more than one heat treated furnace load or a maximum of 50 bodies from a continuous heat treatment process austenitized in the same furnace, at the same time, quenched in the same manner, and tempered in the same tempering furnace at the same time. If all the heat treatment lots comprising the first 1000 bodies meet the requirements of 3.2.2, the heat treatment lot size may be increased. Failure of any of the heat treatment lots comprising the first 1000 bodies shall be cause for maintaining the lotting concept based on a single furnace load, or a maximum of 50 bodies from a continuous process until 1000 bodies have been produced without a failing lot, after which the increased lot size may be reverted to. The increased lot size shall contain bodies from not more than one heat of steel. The heat treatment lot shall be made up of furnace loads as defined above, from the same austenitizing and tempering furnaces. The heat treatment lot thus constituted shall be the basis for selecting the samples for the tension test hereinafter prescribed. In the event of lot rejection of the increased lot size, the original qualifying criteria of producing 1000 bodies consecutively that meet

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4.3.1.2.1 (Cont'd)

the requirements of 3.2.2 shall be reverted to. If the cause of the failure can be determined and corrected, the qualification number of 1000 may be reduced at the discretion of the Administrative Contracting Officer.

4.3.1.2.2 Windshield forgings, base plugs and fuze lock cups.-Heat treatment lots shall contain windshield forgings, base plugs or fuze lock cups from not more than one heat of steel that have been treated in the same heat treatment equipment, at the same temperature, in one unchanged process. The heat treatment lot thus constituted shall be the basis for selecting the samples for the tension test hereinafter prescribed.

4.3.1.3 Stress relieved lots.-Stress relieved lots shall contain cold shaped windshields from not more than one heat of steel that have been cold shaped by the same process and stress relieved in the same furnace at the same temperature for the same length of time. The stress relieved lot thus constituted shall be the basis for selecting the samples for the tension test hereinafter prescribed.

4.3.1.4 Projectile lots.-Bodies from heat treated lots that have complied with requirements of 3.2.2 and windshields from heat treated or stress relieved lots that have complied with the mechanical properties requirements of the applicable drawing, or base plugs and fuze lock cups from heat treated lots that have complied with the mechanical properties requirements of the applicable drawings may be re-grouped for assembly into projectile lots. A projectile lot shall consist of projectile assemblies produced by one manufacturer, under one contract, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision.

4.3.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. An AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect shall be used. All nonconforming material shall be rejected. The roughness comparison specimens prescribed by ASA-B46.1 shall be used as a basis for roughness determination.

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4.3.2.1 Cap, Windshield, Drawing C10534413, covering a detail of Drawing F10534407.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Runout of rear outside diameter with pitch diameter of rear thread and including rear face	Gage
102. Runout of forward outside diameter with pitch diameter of rear thread and including rear face	Gage
103. True position of cavity with pitch diameter of rear thread	Gage
104. True position of pitch diameter of forward thread with cavity	Gage
105. Pitch diameter of rear thread, max.	Gage
106. Minor diameter of rear thread, max.	Gage
107. Pitch diameter of forward thread, max.	Gage
108. Minor diameter of forward thread, max.	Gage
109. Length to basic diameter, inside rear taper	Gage
110. Length to outside forward basic diameter	Gage
111. Length to outside rear basic diameter	Gage
112. Length to basic diameter on inside forward basic angle	Gage
113. Length of effective thread, min.	Gage
114. Flat, front end, max.	Gage
115. Angle, inside rear taper	Gage
116. Profile of basic angle, inside forward end	Gage
117. Thread(s) damaged or not full	Visual
118. Metal defective	Visual
MINOR	
201. Pitch diameter of forward thread, min.	Gage
202. Minor diameter of forward thread, min.	Gage
203. Pitch diameter of rear thread, min.	Gage
204. Minor diameter of rear thread, min.	Gage
205. Diameter of cavity	Gage
206. Depth of thread cavity	Gage
207. Length, total	Gage

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4.3.2.1 (Cont'd)

	Defect	Method of Inspection
208.	Radii or chamfers missing or incorrect	Visual (See 6.2)
209.	Surface finish improper	Visual (See 6.2)
210.	Protective coating inadequate or defective	Visual
211.	Burr	Visual
212.	Foreign matter	Visual

4.3.2.2 Eyelet Terminal, Drawing C10534419, covering a detail of Drawing F10534407.

CRITICAL

None defined

MAJOR

101.	True position of outside diameter rear with hole	Gage
102.	Protective coating inadequate or damaged	Visual

MINOR

201.	Diameter of knurl	Gage
202.	Diameter, outside rear	Gage
203.	Width of wire groove, max.	Gage
204.	Burr	Visual

4.3.2.3 Insulator, Nose, Drawing C10534420, covering a detail of Drawing F10534407.

CRITICAL

None defined

MAJOR

101.	Diameter of hole	Gage

MINOR

None defined

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4.3.2.4 Wire and Eyelet Assembly, Drawing C10534418, covering a detail of Drawing F10534407.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Metal defective (eyelet terminal)	Visual
102. Burr (eyelet terminal)	Visual
103. Corrosive matter (area of tinned wire, if applicable)	Visual
104. Wire strands cut or damaged	Visual
105. Improper assembly	Visual
106. Foreign matter	Visual
MINOR	
None defined	

4.3.2.5 Insulator assembly, Drawing B10534417, covering a detail of Drawing F10534407.

CRITICAL	
None defined	
MAJOR	
101. Diameter of insulator, min.	Gage
102. Diameter of hole in eyelet terminal	Gage
103. Width across tabs, min.	Gage
104. Width of tabs, min.	Gage
105. Shrinkage at central portion of sidewall, max.	Gage
106. Shrinkage at central portion of tabs, max.	Gage
107. Components damaged or defective	Visual
108. Improper assembly	Visual
MINOR	
201. Length of insulator	Gage
202. Radii or chamfers missing or incorrect	Visual (See 6.2)

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4.3.2.5 (Cont'd)

	Defect	Method of Inspection
203.	Burr	Visual
204.	Foreign matter	Visual

4.3.2.6 Windshield, Drawing D10534412, covering a detail of Drawing F10534407.

CRITICAL

None defined

MAJOR

101.	True position of insulator cavity with major diameter of windshield cap thread	Gage
102.	Straightness of ogive	Gage
103.	Pitch diameter of locking ring thread, max.	Gage
104.	Minor diameter of locking ring thread, max.	Gage
105.	Diameter of insulator cavity, max.	Gage
106.	Depth of insulator cavity	Gage
107.	Length of effective locking ring thread, min.	Gage
108.	Length of locking ring thread, max.	Gage
109.	Length to inside basic diameter	Gage
110.	Length to forward outside basic diameter	Gage
111.	Length to rear outside basic diameter	Gage
112.	Wall thickness	Gage
113.	Thread(s) damaged or not full	Visual
114.	Metal defective	Visual

MINOR

201.	Diameter of thread counterbore	Gage
202.	Diameter of cap thread pilot	Gage
203.	Diameter of inside forward of internal thread	Gage
204.	Diameter outside, rear end	Gage
205.	Diameter, rear of insulator cavity	Gage
206.	Depth of thread counterbore	Gage
207.	Length from rear face to cap seat	Gage
208.	Length from front face to cap seat	Gage
209.	Length of cap thread pilot	Gage
210.	Width of insulator tab slots, max.	Gage
211.	Width across insulator tab slots, max.	Gage
212.	Radii or chamfers missing or incorrect	Visual (See 6.2)
213.	Surface finish, improper	Visual (See 6.2)

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4.3.2.6 (Cont'd)

Defect	Method of Inspection
214. Burr	Visual
215. Foreign matter	Visual

4.3.2.7 Ring, Locking, Drawing D10534411, covering a detail of Drawing F10534407.

CRITICAL

None defined

MAJOR

101. Rear face not perpendicular with pitch diameter of body thread	Gage
102. Pitch diameter of windshield thread, min.	Gage
103. Major diameter of windshield thread, min.	Gage
104. Pitch diameter of body thread, min.	Gage
105. Major diameter of body thread, min.	Gage
106. Diameter of body thread pilot	Gage
107. Length of effective windshield thread, min.	Gage
108. Length from rear face to front of body thread	Gage
109. Thread(s) damaged or not full	Visual
110. Protective coating inadequate or damaged	Visual
111. Metal defective	Visual

MINOR

201. Diameter, inside	Gage
202. Length of body thread pilot	Gage
203. Length, total	Gage
204. Presence and location of wrench slots or holes	Visual (See 6.2)
205. Surface finish improper	Visual (See 6.2)
206. Radii or chamfers missing or incorrect	Visual (See 6.2)
207. Burr	Visual
208. Foreign matter	Visual

4.3.2.8 Body, Drawing F10534409, covering a detail of Drawing F10534407.

CRITICAL

1. Metal defective	Visual

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4.3.2.8 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Runout of rotating band seat with bourrelets	Gage
102. Diameter of rotating band seat	Gage
103. Depth of scores in band seat ribs	Visual (See 6.2)
MINOR	
201. Location of rotating band seat	Gage
202. Width of rotating band seat	Gage
203. Burr	Visual
204. Foreign matter	Visual
205. Evidence of shot or sandblasting of band seat after scoring	Visual

4.3.2.9 Plug, Base, Drawing C10542955, covering a detail of Drawing F10534407.

CRITICAL	
1. Metal defective	Visual
MAJOR	
101. True position of tracer cavity with pitch diameter of external thread and including front face of shoulder	Gage
102. True position of tracer cavity with minor diameter of internal thread	Gage
103. Perpendicularity of front face of shoulder with pitch diameter of external thread	Gage
104. Flatness of shoulder	Gage
105. Pitch diameter of external thread, min.	Gage
106. Major diameter of external thread, min.	Gage
107. Pitch diameter of internal thread, max.	Gage
108. Minor diameter of internal thread, max.	Gage
109. Length from flange to front face	Gage
110. Length of effective internal thread, min.	Gage
111. Length of effective external thread, min.	Gage
112. Length of forward pilot diameter	Gage
113. Diameter of tracer cavity	Gage
114. Depth of internal shoulder	Gage

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4.3.2.9 (Cont'd)

	Defect	Method of Inspection
115.	Depth of tracer cavity	Gage
116.	Thread(s) damaged or not full	Visual

MINOR

201.	True position of wrench flats with outside diameter	Gage
202.	Pitch diameter of external thread, max.	Gage
203.	Major diameter of external thread, max.	Gage
204.	Pitch diameter of internal thread, min.	Gage
205.	Minor diameter of internal thread, min.	Gage
206.	Diameter, outside	Gage
207.	Diameter of forward pilot diameter	Gage
208.	Length, total	Gage
209.	Width across wrench flats	Gage
210.	Surface finish improper	Visual (See 6.2)
211.	Radii or chamfers missing or incorrect	Visual (See 6.2)
212.	Protective coating inadequate or defective	Visual
213.	Burr	Visual
214.	Foreign matter	Visual

4.3.2.10 Cup, Fuze Lock, Drawing C10542956, covering a detail of Drawing F10534407.

CRITICAL

1.	Metal defective	Visual

MAJOR

101.	Runout of internal pitch diameter with pitch diameter of external thread and including forward surface and bottom of inside cavity	Gage
102.	Perpendicularity of forward surface with pitch diameter of external thread	Gage
103.	Flatness of forward surface	Gage
104.	Pitch diameter of external thread, min.	Gage
105.	Major diameter of external thread, min.	Gage
106.	Pitch diameter of internal thread, max.	Gage
107.	Minor diameter of internal thread, max.	Gage
108.	Length of effective internal thread, min.	Gage
109.	Length, total	Gage

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Defect	Method of Inspection
110. Depth of counterbore	Gage
111. Thread(s) damaged or not full	Visual
MINOR	
201. Pitch diameter of external thread, max.	Gage
202. Major diameter of external thread, max.	Gage
203. Pitch diameter of internal thread, min.	Gage
204. Minor diameter of internal thread, min.	Gage
205. Diameter of counterbore	Gage
206. Wall thickness, forward end	Gage
207. Presence and location of wrench holes	Visual (See 6.2)
208. Surface finish improper	Visual (See 6.2)
209. Radii or chamfers missing or incorrect	Visual (See 6.2)
210. Protective coating inadequate or defective	Visual
211. Burr	Visual
212. Foreign matter	Visual

4.3.2.11 Ring, Case Locking, Drawing C10534384, covering a detail of Drawing F10534407.

CRITICAL	
None defined	
MAJOR	
101. Pitch diameter of thread, max.	Gage
102. Minor diameter of thread, max.	Gage
103. Thread(s) damaged or not full	Visual
104. Metal defective	Visual
MINOR	
201. Pitch diameter of thread, min.	Gage
202. Minor diameter of thread, min.	Gage
203. Diameter outside	Gage
204. Depth of flat	Gage
205. Location of flat	Gage
206. Width of ring	Gage
207. Width across flats	Gage
208. Radii or chamfers missing or incorrect	Visual (See 6.2)
209. Surface finish improper	Visual (See 6.2)
210. Protective coating inadequate or damaged	Visual
211. Burr	Visual
212. Foreign matter	Visual

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4.3.2.12 Body and Band Assembly, Drawing F10534408, covering a detail of Drawing F10534407.

Defect	Method of Inspection
CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Runout of forward cavity with bourrelets	Gage
102. Runout of fuze hole with bourrelets	Gage
103. Runout of pitch diameter of fuze lock cup thread with bourrelets	Gage
104. Runout of body with bourrelets	Gage
105. Runout of rotating band with bourrelets	Gage
106. True position of major diameter of case locking ring thread with pitch diameter of fuze lock cup thread	Gage
107. True position of diameter forward of fuze lock cup thread undercut with fuze hole (if applicable)	Gage
108. Perpendicularity of rear face with bourrelets	Gage
109. Perpendicularity of fuze seat with bourrelets	Gage
110. Perpendicularity of base plug seat with pitch diameter of fuze lock cup thread	Gage
111. Flatness of base plug seat	Gage
112. Pitch diameter of locking ring thread, max.	Gage
113. Minor diameter of locking ring thread, max.	Gage
114. Diameter of fluted liner seat	Gage
115. Diameter of forward cavity	Gage
116. Diameter of bourrelets, min.	Gage
117. Diameter of rotating band	Gage
118. Diameter of fuze hole	Gage
119. Diameter of body	Gage
120. Diameter forward of fuze lock cup thread undercut (if applicable)	Gage
121. Location of rotating band	Gage
122. Length, total	Gage
123. Length to basic diameter on rotating band	Gage
124. Depth from front face to forward basic diameter, inside forward cavity	Gage
125. Depth from front face to fluted liner seat	Gage
126. Depth from rear face of body to rear basic diameter, inside forward cavity	Gage

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4.3.2.12 (Cont'd)

Defect	Method of Inspection
127. Depth from rear face of body to small basic diameter, inside rear cavity	Gage
128. Depth from rear face of body to large basic diameter, inside rear cavity	Gage
129. Length of effective locking ring thread, min.	Gage
130. Width of rotating band	Gage
MINOR	
201. Diameter of locking ring thread counterbore	Gage
202. Diameter of conduit hole	Gage
203. Diameter of fuze lock cup thread undercut (if applicable)	Gage
204. Diameter forward of fuze lock cup thread, min. (if applicable)	Gage
205. Width of fuze lock cup thread undercut, max. (if applicable)	Gage
206. Width of undercut front and rear of rotating band, max. (if applicable)	Visual (See 6.2)
207. Length of outside basic angle from front face	Gage
208. Width of slot	Gage
209. Depth from rear face of body to bottom of rear cavity	Gage
210. Depth of slot	Gage
211. Depth of undercut front and rear of rotating band, max. (if applicable)	Visual (See 6.2)
212. Depth from base plug seat to bottom or rear cavity	Gage
213. Depth of locking ring thread counterbore, max.	Gage
214. Location of slot	Gage
215. Length of locking ring thread, max.	Gage
216. Length from rear face to front of rear bourrelet	Gage
217. Length from front face to rear of forward bourrelet	Gage
218. Length from base plug seat to fuze lock cup seat (if applicable)	Gage
219. Length from base plug seat to fuze seat	Gage

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4.3.2.12 (Cont'd)

	Defect	Method of Inspection
220.	Depth of conduit hole (if applicable)	Gage
221.	Location of conduit hole	Gage
222.	Radii or chamfers missing or incorrect	Visual (See 6.2)
223.	Surface finish, improper	Visual (See 6.2)
224.	Burr	Visual
225.	Foreign matter	Visual

4.3.2.13 Metal Parts Assembly, Drawing F10534407, prior to painting.

CRITICAL		
1.	Gap between windshield and body	Gage
MAJOR		
101.	Insecure assembly (locking ring to body)	Torque Wrench (See 4.3.3.10)
102.	Insecure assembly (windshield to locking ring)	Torque Wrench (See 4.3.3.10)
103.	Runout of pitch diameter of windshield cap threads with bourrelets, and including windshield cap seat	Gage
104.	Runout of forward basic diameter of windshield with bourrelets	Gage
105.	Runout of rear basic diameter of windshield with bourrelets	Gage
106.	Runout at rear of fluted liner with bourrelets	Gage
107.	Length total, max.	Gage
108.	Rotation of fluted liner	Tactile/Visual
MINOR		
201.	Marking missing, incorrect or illegible	Visual
202.	Burr	Visual
203.	Foreign matter	Visual

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4.3.2.14 Metal Parts Assembly, Drawing F10534407 (after painting).

	Defect	Method of Inspection
CRITICAL		
	None defined	
MAJOR		
101.	Weight	Approved Scale
102.	Pitch diameter of windshield cap thread, min.	Gage
103.	Pitch diameter of case locking ring thread, min.	Gage
104.	Pitch diameter of fuze lock cup thread, max.	Gage
105.	Major diameter of windshield cap thread, min.	Gage
106.	Major diameter of case locking ring thread, min.	Gage
107.	Minor diameter of fuze lock cup thread, max.	Gage
108.	Diameter of bourrelets, max.	Gage
109.	Length of effective windshield cap thread, min.	Gage
110.	Length of effective case locking ring thread, min.	Gage
111.	Length of effective fuze lock cup thread, min. (if applicable)	Gage
112.	Thread(s) damaged or not full	Visual
113.	Rotating band damaged	Visual
114.	Alignment of conduit between fluted liner and body	Visual
115.	Fuze wire not properly aligned and assembled in conduit	Visual/Manual
116.	Foreign matter in charge cavity	Visual
MINOR		
201.	Pitch diameter of windshield cap thread, max.	Gage
202.	Pitch diameter of case locking ring thread,	Gage
203.	Pitch diameter of fuze lock cup thread, min.	Gage
204.	Major diameter of windshield cap thread, max.	Gage
205.	Major diameter of case locking ring thread, max.	Gage
206.	Minor diameter of fuze lock cup thread, min.	Gage
207.	Location of wire terminal	Measure
208.	Protrusion of conduit from base plug seat	Measure
209.	Paint not dry	Tactile
210.	Paint on thread(s)	Visual
211.	Protective coating inadequate or damaged	Visual
212.	Marking missing, incorrect or illegible	Visual

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4.3.2.14 (Cont'd)

	Defect	Method of Inspection
213.	Burr	Visual
214.	Foreign matter (except as otherwise specified)	Visual

4.3.2.15 Conduit, Drawing B10534416, covering a detail of Drawing F10534407.

CRITICAL

None defined

MAJOR

101.	Presence and size of flared end	Visual (See 6.2)
102.	Diameter, outside	Gage
103.	Material defective	Visual

MINOR

201.	Diameter, inside	Gage

4.3.2.16 Plug and Disc Assembly, Drawing B8595462, covering a detail of Drawing F10534407.

CRITICAL

None defined

MAJOR

101.	True position of inside diameter with major diameter of thread	Gage
102.	Pitch diameter of thread, min.	Gage
103.	Major diameter of thread, min.	Gage
104.	Diameter of thread chamfer, min.	Gage
105.	Width of threaded diameter	Gage
106.	Thickness of disc	Gage
107.	Thread(s) damaged or not full	Visual
108.	Improper assembly	Visual
109.	Metal defective	Visual

MINOR

201.	Pitch diameter of thread, max.	Gage
202.	Major diameter of thread, max.	Gage

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4.3.2.16 (Cont'd)

	Defect	Method of Inspection
203.	Diameter of thru hole	Gage
204.	Presence and location of wrench holes	Visual (See 6.2)
205.	Surface finish improper	Visual (See 6.2)
206.	Burr	Visual
207.	Foreign matter	Visual
208.	Protective coating inadequate or damaged	Visual

4.3.2.17 Plug, Nose, Drawing B11738337, covering a detail of Drawing F10534407.

CRITICAL

None defined

MAJOR

101.	Pitch diameter of thread, min.	Gage
102.	Major diameter of thread, min.	Gage
103.	Diameter of thread chamfer	Gage
104.	Thickness of body	Gage

MINOR

201.	Width and depth of slot	Visual (See 6.2)
202.	Surface finish improper	Visual (See 6.2)
203.	Burr	Visual
204.	Protective coating inadequate or damaged	Visual
205.	Foreign matter	Visual

4.3.2.18 Cap, Windshield Assembly, Drawing C11738336, covering a detail of Drawing F10534407.

CRITICAL

None defined

MAJOR

101.	Nose plug flush or below front face	Gage
102.	Insecure assembly, min.	Torque Wrench (See 4.3.3.10)

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4.3.2.18 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Burr	Visual
202. Foreign matter	Visual

4.3.2.19 Pallet, Drawing F8889328.

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Pallet damaged (unserviceable)	Visual
202. Sleeve(s) missing	Visual
203. Improperly packed	Visual
204. Strapping missing, loose or broken	Visual/Manual
205. Marking missing, incorrect or illegible	Visual

4.3.2.20 Packed Carton (prior to sealing).

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Improper assembly	Visual
202. Support missing or damaged	Visual
203. Separator missing or damaged	Visual
204. Incorrect number in carton	Visual

4.3.2.21 Packed Carton (after sealing).

CRITICAL	
None defined	

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4.3.2.21 (Cont'd)

Defect	Method of Inspection
MAJOR	
None defined	
MINOR	
201. Carton damaged, cut or gauged through all layers	Visual
202. Carton not properly sealed	Visual
203. Contents loose	Visual
204. Marking missing, incorrect or illegible	Visual

4.3.3 Inspection testing.-The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Mechanical properties.

4.3.3.1.1 Hardness.

4.3.3.1.1.1 Heat treatment lots.-Each body, windshield, base plug or fuze lock cup in the heat treatment lot (as defined by 4.3.1.2) shall be subjected to a hardness test to determine components to be selected for the tension test. Failure of any base plug or fuze lock cup to meet the hardness requirement of the applicable drawing shall be cause for rejection of the component.

4.3.3.1.1.2 Stress relieved lots.-Twenty-five cold shaped windshields, from the first batch of each stress relieved lot (as defined by 4.3.1.3) processed through the stress relief furnace and five windshields from each one hour production thereafter for the balance of the stress relieved lot shall be subjected to a hardness test to assess uniformity of the process. Records of hardness results shall be maintained.

4.3.3.1.2 Tension test.

4.3.3.1.2.1 Heat treatment lots.

4.3.3.1.2.1.1 Bodies.-Two bodies, one of the hardest and one of the softest as determined by 4.3.3.1.1.1 shall be subjected to this test. Failure of any test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the lot.

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4.3.3.1.2.1.2 Windshield, base plug and fuze lock cup.-Two windshields, two base plugs and two fuze lock cups, one of the hardest and one of the softest as determined by 4.3.3.1.1.1 shall be subjected to this test. Failure of any tension test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the lot. At the option of the supplier, a production control sample may be utilized for these components to establish a hardness range. When a production control sample is used, the hardness range thus established shall be maintained throughout the heat treatment range. The size of the sample shall be determined by the supplier. Windshields, base plugs or fuze lock cups, which fall outside of the established range may be reheat treated to fall within the range or new mechanical properties may be determined from newly selected components which will represent the true extremes of hardness in the heat treatment lot.

4.3.3.1.2.2 Stress relieved lots.-Two windshields from each stress relieved lot shall be selected at random and subjected to this test to determine compliance with mechanical properties requirements of the applicable drawing. Failure of any tension test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the lot.

4.3.3.2 Decarburization.-One body from each heat treatment lot of the lowest hardness as determined by 4.3.3.1.1.1 shall be subjected to this test. Failure of any specimen to comply with 3.2.2.1 shall be cause for rejection of the heat treatment lot.

4.3.3.3 Magnetic particle inspection.

4.3.3.3.1 Body prior to banding.-After heat treatment and machining, including band seat, prior to banding, each body shall be subjected to a magnetic particle inspection. Failure of any body to comply with the requirement of 3.4.2 shall be cause for rejection of the component.

4.3.3.3.2 Body after banding.-After banding and final machining, prior to painting, each body and band assembly shall be subjected to a magnetic particle inspection. Failure of any body and band assembly to comply with requirements of 3.4.2 shall be cause for rejection of the assembly.

4.3.3.4 Rotating band.

4.3.3.4.1 Bandseating.-Two body and band assemblies from each lot, prior to painting, shall be subjected to this test to determine compliance with the requirements of the applicable drawing. Failure of either assembly to comply shall be cause for rejection of the lot.

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4.3.3.4.2 Density.-The samples provided by 4.3.3.4.1 after examination for seating shall be subjected to this test to determine compliance with requirements of the applicable drawing. Failure of any specimen to comply shall by cause for rejection of the lot.

4.3.3.5 Cleaning and application of protective coating.-The sample size shall be as specified in the applicable specification with the exception that the sample size for the corrosive resistance test for those components requiring protective finish in accordance with MIL-C-60539, Type II, Class 1 shall be selected in accordance with MIL-STD-105, Inspection Level S-2. The acceptance criteria shall be as specified in the applicable specification. Failure to comply with the requirements of the applicable drawings and specifications shall be cause for taking action prescribed in the applicable specification.

4.3.3.6 Strength of wire assembly.-Twenty-five insulator assemblies from each lot shall be subjected to a pull test to determine compliance with requirements of the applicable drawing. Failure of any assembly to comply shall be cause for rejection of the lot.

4.3.3.7 Fuze wire terminal.-The fuze wire terminals of 25 metal parts assemblies from each lot shall be subjected to a pull test to determine compliance with requirments of the applicable drawing. Failure of any assembly to comply shall be cause for rejection of the lot.

4.3.3.8 Electrical.

4.3.3.8.1 Electrical breakdown.-Each metal parts assembly shall be subjected to this test to determine compliance with requirements of the applicable drawing. Failure of any assembly to comply shall be cause for rejection of the assembly.

4.3.3.8.2 Electrical continuity.-Each metal parts assembly shall be subjected to this test to determine compliance with requirements of the applicable drawing. Failure of any assembly to comply shall be cause for rejection of the assembly.

4.3.3.8.3 Electrical resistance.-Each metal parts assembly shall be subjected to this test to determine compliance with requirements of the applicable drawing. Failure of any assembly to comply shall be cause for rejection of the assembly.

4.3.3.9 Air pressure.-Each assembly consisting of body and band assembly, locking ring, fluted liner and conduit shall be subjected to an air pressure test to determine compliance with the requirements of the applicable drawing. Failure of any assembly to comply shall be cause for rejection of the assembly.

4.3.3.10 Torque test.

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4.3.3.10.1 Locking ring to body.-Locking ring and body assemblies shall be subjected to this test to determine compliance with the requirements of the applicable drawing. Failure of the sample to comply with the criteria of the sampling plan as evidenced by a perceptible movement shall be cause for rejection of the lot.

4.3.3.10.2 Windshield to locking ring.-Windshield and locking ring assemblies shall be subjected to this test to determine compliance with the requirements of the applicable drawing. Failure of the sample to comply with the criteria of the sampling plan as evidenced by any perceptible movement shall be cause for rejection of the lot.

4.3.3.10.3 Nose plug to windshield cap.-Nose plug and windshield cap assemblies shall be subjected to this test to determine compliance with the requirements of the applicable drawing. Failure of the sample to comply with the criteria of the sampling plan as evidenced by any perceptible movement shall be cause for rejection of the lot.

4.3.3.11 Macroetch.-Two forgings randomly selected from each forging lot of fuze lock cups shall be tested to determine compliance with drawing. Failure of either specimen to comply, as evidenced by improper direction of flow lines, shall be cause for rejection of the lot. For the purpose of this test a forging lot shall contain forgings from not more than one heat of steel and represent a maximum of 3,000 forgings.

4.3.3.12 Tool control dimension.-Dimension designated as tool control type dimensions shall be gaged at the beginning of each eight hours production and whenever tooling is replaced. Where destruction of the component is necessary to inspect these dimensions, measurement of the tool may be substituted provided the contractor has established correlation between the tool dimension and the component dimension prior to the start of production. The contractor shall be responsible for maintaining adequate surveillance to assure that drawing requirements are being met throughout the life of the tool.

4.3.4 Proving ground functional test.-The following proving ground tests will be performed by the Government.

4.3.4.1 First article sample.

4.3.4.1.1 Initial production sample.-The quantities listed in Table I shall be ballistically tested in the phase indicated.

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4.3.4.1.1 (Cont'd)

TABLE I

Test Phase	Quantity
Excess pressure	15
Service pressure	15
Fragmentation	2

4.3.4.2 Production lot sample.-The quantities listed in Table II shall be ballistically tested in phases indicated.

TABLE II

Test Phase	Quantity
Excess pressure	10
Service pressure	10

4.3.4.3 Excess pressure.-Samples provided in accordance with Tables I and II will be subjected to this test. Failure of any projectile assembly to comply with 3.3, shall be cause for rejection of the initial production sample or the production lot represented by the sample.

4.3.4.4 Service pressure.-Samples provided in accordance with Tables I and II will be subjected to this test. Failure of any projectile assembly to comply with 3.3, paragraph "b" shall be cause for rejection of the initial production sample or the production lot represented by the sample.

4.3.4.5 Fragmentation.-Metal parts assemblies supplied in accordance with Table I shall be subjected to this test. Testing will be performed for information only.

4.3.5 Inspection equipment.-Index of Inspection Equipment Lists LI 10534407 identify the inspection equipment required to perform the examination and test prescribed herein. Equipment design shall be in accordance with code designation on the applicable EL (Reference is made to inspection equipment list for definition of coding format).

4.4 Test methods and procedures.

4.4.1 Mechanical properties.

4.4.1.1 Hardness.-After final heat treatment or stress relief and before final machining, the hardness test shall be performed on the body at

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4.4.1.1 (Cont'd)

the location of the rotating band seat and on the rear outside diameter of the windshield. The tests shall be made using the procedure for performing this test as prescribed by ASTM-E-18.

4.4.1.2 Tension test.

4.4.1.2.1 Body.-From the components selected in accordance with 4.3.3.1.2, one tensile test specimen shall be taken longitudinally from midpoint of the body sidewall and one tensile test specimen shall be taken in a tangentially transverse direction and located approximately 2 inches from the axis of the body from the central portion of the base web area. The specimens shall conform to the largest obtainable Standard "R" type specimens available as described by ASTM-E-8. Yield strength shall be determined by the offset method, utilizing the procedures prescribed by ASTM-E-8.

4.4.1.2.2 Windshield.-From the components selected in accordance with 4.3.3.1.2, two full length specimens shall be taken longitudinally 180 degrees apart. The specimens shall conform to the largest obtainable standard "T" type specimen as described by ASTM-E-8. Yield strength shall be determined by the offset method using a limiting set of 0.1 percent offset, utilizing the procedures prescribed by ASTM-E-8. At the option of the supplier, a standard "R" type specimen or "F" type specimen, as described by ASTM-E-8, may be utilized for making yield strength and elongation determinations on the heat treated or stress relieved windshield prior to machining, provided the largest obtainable specimens are no smaller than a standard R3 specimen or F2 specimen. The largest obtainable standard "T" type specimen will be used by the Government for initial production lot verification and in any subsequent Government testing of the windshield as may be necessary.

4.4.1.2.3 Base plug.-From the components selected in accordance with 4.3.3.1.2 a tangentially transverse specimen shall be taken. Its centerline shall be located approximately 5/8 inch from the axis of the base plug and 1/2 inch from the rear end. The specimen shall conform to the largest obtainable standard "R" type specimen as described by ASTM-E-8. Yield strength shall be determined by the offset method utilizing the procedures prescribed by ASTM-E-8.

4.4.1.2.4 Fuze lock cup.-From the components selected in accordance with 4.3.3.1.2 a transverse specimen shall be removed from the front end of the fuze lock cup. The specimen shall conform to the largest obtainable standard "R" type specimen as described by ASTM-E-8. Yield strength shall be determined by the offset method utilizing the procedures prescribed by ASTM-E-8.

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4.4.1.3 Decarburization.-From one body, selected in accordance with 4.3.3.2, one metallographic specimen shall be sectioned from the side wall approximately 3 inches from the front end. The specimen shall represent the full cross section of the finish machined body. The specimen shall be suitably etched with an etchant, such as 5% picrol and then examined at 100 diameters to determine compliance with 3.2.2.1. The zone of decarburization shall include the total depth of complete and partial decarburization.

4.4.2 Magnetic particle inspection.

4.4.2.1 Body.-The procedure for performing this test shall be as specified in MIL-I-6868 for wet fluorescent method, indirect magnetization using either the continuous method or the residual method. The inspection procedure both before and after banding shall include two inspections: Once after inducing a circular magnetic field in the component and once after inducing a longitudinal magnetic field in the component. The magnetizing current used to induce magnetism in the component shall be between 2500 and 2800 amperes. The component shall be examined for indications of discontinuity on all exterior and interior surfaces. Bodies prior to banding that have been inspected and accepted shall be stamped with a letter "M" in the area between the bourrelets. Body and band assemblies that have been inspected and accepted shall be stamped with the letter "F" in the same area. The stamped letters "M" and "F" shall be oriented in the same direction as the nomenclature stamp.

4.4.2.2 Suspension.-For the inspections specified in 4.4.2.1, the solid content of the suspension shall be maintained at 0.2 to 0.4 milliliters (ml) of solids per 100 ml of suspension when oil is used or 0.15 to 0.35 ml of solids per 100 ml of suspension when water is used. The suspension shall be checked at the start of each day's production and at intervals of not over four hours of continuous production. When performing the test for solid content of the suspension, if it is determined that it does not comply, all components that have been tested and accepted since the last satisfactory test was performed shall be retested after the suspension has been brought to the proper level. The magnetic particle test shall be performed on equipment that complies with the requirements of PA-PD-792. The operator(s) of the equipment and the inspector(s) making the determination shall be certified in accordance with MIL-STD-410.

4.4.3 Rotating band.

4.4.3.1 Band seating.-The body and band assemblies provided by 4.3.3.4 shall be carefully sectionalized with a wet abrasive cutoff wheel, so as to produce 4 segments of 90 degrees each. Visual examination of the exposed surfaces of the rotating band shall then be conducted to determine compliance with the band seating requirements of the applicable drawing.

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4.4.3.2 Density.-After examination for band seating as prescribed in 4.4.3.1, two specimens from each body and band assembly spaced approximately 180 degrees apart, shall be removed from the rotating bands. The specimens shall be the full width and thickness of the band and approximately one inch long. The specimens shall be sawed by hand to the specified size. With the exception of removing the burrs, no further machining shall be performed on the specimens. After weighing the specimens, the wax shall be removed by either one of the two methods described in "a" and "b".

a. Place specimens in a dry hydrogen atmosphere furnace and hold at 1400 degrees F. for one hour. Remove specimens and weigh. Return specimens to the furnance and hold at 1400 degrees F. for 30 minutes. This procedure with the half hour heating intervals shall be repeated until the change in weight is less than .05%.

b. A Soxhlet apparatus of suitable size using either toluene or trichloroethylene may be used for extracting the wax. The extraction time shall be a minimum of 18 hours. The specimens shall then be dried for 1½ hours in an oven at 210 degrees F. and then weighed. This procedure with 1½ hour extraction intervals shall be repeated until the change in weight is less than .05%.

Following wax removal and weighing, the specimens shall be impregnated with micro-crystalline wax. This shall be accomplished by immersing the specimen in molten micro-crystalline wax held at a temperature of 180 degrees F. to 200 degrees F. The pressure shall then be reduced by a suitable evacuating method to not more than 2 inches of mercury and held for 30 minutes. The pressure shall then be slowly increased (approximately 30 seconds) to atmospheric pressure after which the specimen shall remain immersed in the molten micro-crystalline wax at 180 degrees F. to 200 degrees F. for approximately one hour.

The density of the specimens shall be calculated as follows:

$$D = \frac{A}{B - C}$$

Where:

D = density in grams per cubic centimeter

A = weight in air of wax free specimen, in grams

B = weight in air of micro-crystalline impregnated specimen, in grams

C = weight of micro-crystalline wax impregnated specimen immersed in water, in grams

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4.4.3.3 Density and wax content of band blank.-The procedure for determining density and wax content shall be as specified in 4.4.3.2 except that from the band blanks selected in accordance with MIL-R-11073, nine specimens shall be sectioned from each band. They shall consist of three sets spaced approximately 120 degrees apart. The approximate size of each specimen shall be one inch long x 1/3 the width of the band x the full thickness of the band. (See Figure 1). The specimens shall be sawed by hand to the specified size. With the exception of removing burrs, no further machining shall be performed on the specimens.

4.4.4 Protective coating.-The procedure for testing the cleaning of components and the application of protective coating shall be in accordance with the criteria for testing as prescribed by the applicable drawing and specification (see 6.3).

4.4.5 Strength of wire assembly.-The samples selected in accordance with 4.3.3.6 shall be placed in a holding device that simulates the actual metal parts assembly and the load applied gradually to the wire in the direction of the wire axis. Assemblies tested shall not be assembled into metal parts assemblies.

4.4.6 Fuze wire terminal.-The samples selected in accordance with 4.3.3.7 shall be tested using a suitable holding device and gradually applying the maximum load required to the wire terminal in the direction of the wire axis.

4.4.7 Electrical.

4.4.7.1 Electrical insulation breakdown.-The direct current shall be applied between the eyelet terminal and rear body threads and the circuit monitored for evidence of insulation breakdown.

4.4.7.2 Electrical continuity.-The voltage shall be applied between the eyelet terminal and the fuze wire terminal and the circuit monitored for evidence of discontinuity.

4.4.7.3 Electrical resistance.-The resistance between the forward end of the windshield and the rear of the body and between the eyelet terminal and fuze wire terminal shall be checked with an ohmmeter capable of performing this test accurately.

4.4.8 Air pressure.-The assembly shall be tested using equipment capable of performing the test accurately and consistently. The device shall be equipped with a gage to visibly determine the pressure applied. The required air pressure shall be applied and held for the minimum time required. The equipment shall be capable of visually indicating a leaking assembly. Assemblies that have been tested and accepted shall be stamped with a letter "A" in the area between the bourrelets. Its orientation shall be the same as the nomenclature stamp.

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4.4.9 Torque test.

4.4.9.1 Locking ring to body.-The assemblies selected in accordance with 4.3.3.10.1 shall be mounted in a suitable holding device in such a manner that the body component is prevented from turning. The applicable torque shall be applied to the locking ring in the direction of assembly. Testing shall be conducted while thread sealant is wet.

4.4.9.2 Windshield to locking ring.-The assemblies selected in accordance with 4.3.3.10.2 shall be mounted in a suitable holding device in such a manner that the body component is prevented from turning. The applicable torque shall be applied to the windshield in the direction of assembly. Testing shall be conducted while thread sealant is wet.

4.4.9.3 Nose plug to windshield cap.-The assemblies selected in accordance with 4.3.3.10.3 shall be mounted in a suitable holding device in such a manner that the windshield cap is prevented from turning. The applicable torque shall be applied to the nose plug in the direction of disassembly. Testing shall be conducted while adhesive is wet.

4.4.10 Macroetch.-The samples selected in accordance with 4.3.3.11 shall be sectionalized in the diametrical plane which corresponds to the rolling direction of the starting material (plate or bar). One section from each forging shall have the sectionalized face ground, polished, and etched to reveal flow lines as prescribed by ASTM-A-317. See Figure II for typical examples of acceptable grain flow patterns.

4.4.11 Proving ground functional tests.

4.4.11.1 Excess pressure.-The metal parts assemblies supplied in accordance with Tables I and II will be assembled into complete rounds, or live loaded and dummy fuzed and fired in a gun fro which the projectile is standard. The propellant charge shall be adjusted to produce the maximum rated pressure (44,200 PSI approximately) in a new gun. Photographic equipment will be set up near the gun muzzle in such a manner as to record the flight of the projectile prior to its impact with the recovery area. The projectiles shall be recovered and examined for the requirements of 3.3.

4.4.11.2 Service pressure.-The metal parts assemblies supplied in accordance with Tables I and II will be assembled into complete rounds, inert or live loaded, dummy fuzed and fired in a gun for which the projectile is standard. The test rounds will be fired alternately with projectiles from a reference lot for comparison. All rounds will have velocities measured and will be fired at a vertical target placed 2000 meters from the muzzle of the gun. Accuracy in terms of maximum spread in inches, both horizontal and vertical and target dispersion in terms of standard deviation in mils shall be recorded for all rounds fired

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4.4.11.2 (Cont'd)

(see 6.1). Velocity and chamber pressure shall also be recorded for all rounds fired (see 6.1). Photographic equipment will be set up near the gun muzzle to record the flight of the projectile prior to its impact with the target. Photographs will be examined for compliance with requirements of 3.3, paragraph "b".

4.4.11.3 Fragmentation.-The metal parts assemblies provided in accordance with Tables I will be placed in a wooden box and buried in a sandpit of suitable size and detonated statically. After detonation, the sand will be screened and the fragments recorded, carefully weighed, segregated by size and classified to determine the effectiveness of the round.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packing.

5.2 Packing.

5.2.1 Level A - Military Pack.-Not applicable.

5.2.2 Level B - Limited Military Pack.-Not applicable.

5.2.3 Level C - Minimum Military Pack.-The component parts for Projectile,152mm: HEAT, M409 Metal Parts shall be packed as follows for shipment to the designated loading facility.

5.2.3.1 Metal Parts Assembly, Part No. 10534407 shall be packed in accordance with Drawing F8889328.

5.2.3.2 Plug and Disc Assembly, Part No. 8595462 shall be packed in accordance with MIL-STD-1169; pack 2500 plug-and-disc assemblies arranged in a 10 x 10 pattern, 25 to a cell, using interlocking separators.

5.2.3.3 Fuze Lock Cup, Part No. C10542956 shall be packed in accordance with MIL-STD-1169. Pack 100 fuze cups arranged in a 5 x 5 pattern, 4 layers, one to a cell, using interlocking separators.

5.2.3.4 Base Plug, Part No. C10542955 shall be packed in accordance with MIL-STD-1169. Pack 50 base plugs arranged in a 5 x 5 pattern, 2 layers, one to a cell, using interlocking separators.

5.2.3.5 Windshield Cap, Part No. C10534413 shall be packed in accordance with MIL-STD-1169. Pack 150 windshield caps arranged in a 6 x 5 pattern, 5 layers, one to a cell, using interlocking separators.

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5.2.3.6 Case Locking Ring, Part No. 10534384 shall be packed in accordance with MIL-STD-1169. Pack 400 case locking rings arranged in a 4 x 5 pattern, 2 layers, 10 to a cell, using interlocking separators.

5.2.3.7 Shipping.-When component parts from more than one lot are shipped as a carload or truckload, each lot must be kept separate and the division between lots clearly indicated.

5.2.3.8 Marking.-The pallet shall be marked in accordance with Drawing F8889328. Cartons shall be marked in accordance with MIL-STD-1169.

5.2.3.9 Data cards.-Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Velocity-accuracy.-Proving Ground tests for velocity and accuracy are not acceptance tests, but the results will be carefully analyzed by the Munitions Command to determine quality of ammunition produced. Evidence of excessive variation in any phase of the Proving Ground tests will result in immediate investigation by the Munitions Command.

6.2 Visual examination qualification.-When compliance with the applicable requirements is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

6.3 Salt spray exposure test as required by Drawing No. F10534407, Metal Parts Assembly, may be performed on rejected components, sections thereof, or cylindrical shapes processed and treated in the same manner as production assemblies.

6.4 The fluted liner, Drawing D10534415, to be used in the metal parts assembly will be furnished by the Government, unless otherwise determined by the procuring activity.

6.5 Ordering data.-Procurement documents should specify the title, number and date of this specification.

6.6 Visual standards.-Visual standards for evaluation of metal defects will be established for each contract after the first month's production.

Custodian:

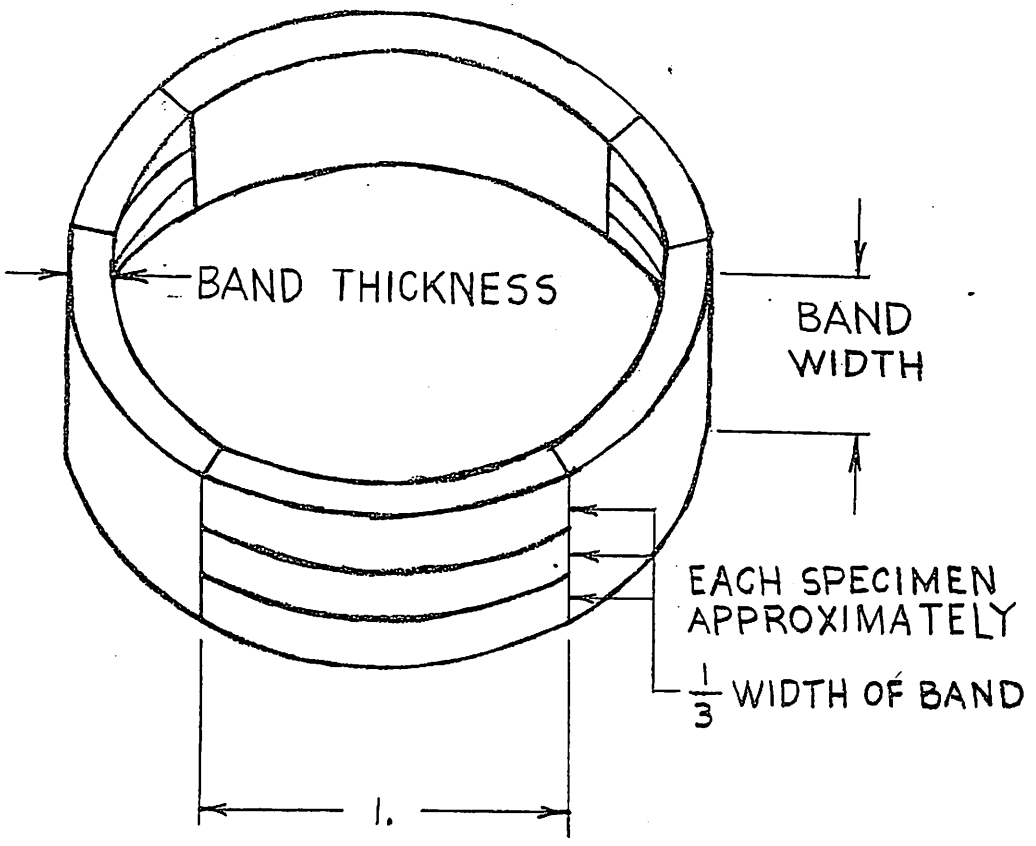
Army - MU

Preparing Activity:

Army - MU(FA)

Project No. 1320- A132

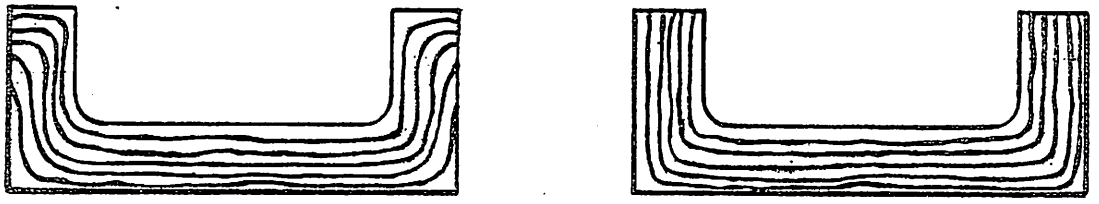
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THREE SETS OF SPECIMENS AS INDICATED, SHALL BE SECTIONED FROM EACH BAND TO BE TESTED, SPACED APPROXIMATELY 120° APART.

Figure I

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ACCEPTABLE GRAIN FLOW PATTERNS FOR FUZE LOCK CUP FORGINGS.

Figure II

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

TRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-P-14825A

ORGANIZATION

CITY AND STATE	CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*	DATE

FORM 1426
1 JAN 66

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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